

Interparticle Spacing, Compressibility and Diffusion

Gases squeeze easily and smells travel because of the spaces between particles and their constant motion.

Spaces between particles

The **interparticle spacing** (gap between particles) is smallest in solids, larger in liquids, and largest in gases. This explains compressibility and diffusion.

	Solid	Liquid	Gas
Interparticle spacing	Very small	Larger	Very large
Can be compressed?	Hardly	Very little	Easily

Compressibility

Block the nozzle of an air-filled syringe and push the plunger — it moves in, because the large spaces between gas particles let them be pushed closer. Try the same with water and the plunger barely moves: liquids (and solids) have little space, so they are almost incompressible.

Diffusion

The spreading of particles of one substance into another on their own is **diffusion**. It happens because particles are **always moving**.

- Fastest in **gases** (e.g. the smell of incense spreads across a room), slower in liquids, slowest/negligible in solids.
- Diffusion is **faster on heating**, because particles move faster.

Why gases fill a container: huge spaces + free, fast-moving particles → gases spread to occupy all available space.

Where students lose marks

Trap 1 — Gases are easily compressed; liquids and solids are not. It's the large spaces in gases that allow compression.

Trap 2 — Diffusion is fastest in gases, slowest in solids. More spacing and free motion → faster diffusion.

Trap 3 — Heating speeds up diffusion, because particles move faster.

Model answers — write them like this

Example A. Why can air in a syringe be compressed but water cannot?

Step 1: Gas particles have large interparticle spaces.

Step 2: Pushing the plunger forces them closer; water's particles are already close, with little space.

∴ Air compresses easily; water is almost incompressible.

Example B. Why does the smell of agarbatti spread quickly across a room?

Step 1: Its particles diffuse through the air.

Step 2: Gas particles move fast and have large spaces, so diffusion is quick.

∴ Fast diffusion of particles in gas spreads the smell.

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